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Automated Rationalization of Subsurface Software Portfolios in Oil & Gas Using Fine-Tuned Sentence-Similarity Transformers

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Abstract

This paper presents an innovative approach for optimizing application landscapes in the oil and gas industry, employing fine-tuned sentence similarity algorithms to rationalize subsurface software portfolios. We developed a unique framework that integrates linguistic analysis with industry-specific nuances, addressing gaps identified in current methodologies. Through a meticulous process of data collection, preprocessing, and algorithm selection, we tailored our approach to effectively process technical language specific to the oil and gas sector. The results from our experiments, which involved rigorous testing and training phases, demonstrated a high degree of accuracy in identifying redundant and overlapping applications, with a ~7% increase in Spearman Correlation post fine-tuning. A case study application in a real-world scenario illustrated the practical benefits of our method, showing a potential of up to ~25% reduction in software redundancies and an improvement in IT resource allocation. Our approach not only enhances the efficiency of software utilization but also sets a new precedent in applying natural language processing techniques for IT portfolio management in the oil and gas industry.

Introduction

Context and Background

Mature IT organizations with legacy applications born on-premise are currently faced with the daunting task of reducing their enterprise sprawl to harness modern applications with architectures usually dependent on cloud architecture. This challenge is exacerbated by the rapid evolution of technology, which has led to an increase in the number of applications used within organizations (Mateus et al., 2021). While these applications offer a variety of functionalities that can improve business processes, they also contribute to a complex and cluttered IT landscape (De Marco et al., 2018).

This complexity not only increases the cost of maintenance but also poses challenges in terms of interoperability and data consistency. In many cases, organizations continue to use legacy applications alongside newer solutions, leading to redundancy and inefficiencies (Ciborowska et al., 2021). This creates a massive gap in knowledge and ability as companies are having their veteran developers replaced with a new, inexperienced workforce (Lee et al., 2022).

Rationalizing the IT application landscape is therefore a critical activity for organizations aiming to streamline operations and reduce costs. However, traditional methods of rationalization often involve manual, time-consuming audits and analyses that are not scalable and are prone to human error (van Deursen et al., 2005).

Natural Language Processing (NLP) techniques, such as sentence similarity algorithms, have recently been employed to automate the process of identifying redundant applications within an organization (Pandita et al., 2021). These algorithms compare the functionalities of different applications and suggest potential candidates for rationalization, thereby aiding decision-makers in the rationalization process.

Problem Statement

Despite the availability of various tools and methodologies for IT rationalization, many organizations still struggle with the effective identification and elimination of redundant applications (Ocampo et al., 2013). Traditional approaches often rely on manual audits and expert reviews, which are time-consuming, costly, and subject to human error (Edwards et al., 2019). Furthermore, these methods often lack the capability to analyze the nuanced functionalities of applications at scale, leading to suboptimal rationalization decisions (Ishmam et al., 2020).

The advent of Natural Language Processing (NLP) offers promising avenues for automating and enhancing the rationalization process. However, there is limited research on the application of NLP techniques, specifically sentence similarity algorithms, in the context of IT rationalization (Gurrapu et al., 2023). This gap in the literature signifies a missed opportunity for organizations to harness advanced computational methods for optimizing their application landscapes.

Objectives and Contributions

Objectives. The primary objective of this research is to introduce a novel, automated approach for rationalizing subsurface software applications within mature IT organizations. Specifically, we aim to:

1. Develop a sentence similarity algorithm that can effectively compare the functionalities of different applications in the subsurface domain.
2. Validate the effectiveness of this algorithm through a case study involving 61 SLB applications and over 200 in-house subsurface applications.
3. Provide actionable insights for IT managers and decision-makers on how to streamline their application landscapes using our proposed methodology (Beulen, 2017).

Contributions. This research makes several significant contributions to both the academic and practical aspects of IT rationalization:

1. **Methodological Contribution:** We present a scalable, automated framework for application rationalization that leverages NLP techniques. This fills a gap in the existing literature and offers a more efficient alternative to traditional methods, which often involve manual audits and are prone to human error (Beulen, 2017; Liao et al., 2018).
2. **Empirical Contribution:** Through our case study, we demonstrate that our sentence similarity algorithm can accurately identify applications with similar functionalities, thereby aiding in the rationalization process.
3. **Practical Contribution:** Our research offers a cost-effective, efficient alternative to traditional rationalization methods, providing actionable insights for IT managers and decision-makers (Beulen, 2017; Liao et al., 2018).

Article Organization

This article is structured to provide a comprehensive understanding of the challenges and solutions associated with application rationalization in mature IT organizations. Section 2 provides an overview of the literature on sentence similarity algorithms and application rationalization. Section 3 outlines the research methodology. The results and discussion of the rationalization outcome of the software are presented in section 4. The paper is concluded in section 5 with a summary of the findings and limitations of the research.

Literature Review

Application rationalization is a critical activity for mature IT organizations aiming to streamline their application portfolios. This process involves the identification, assessment, and consolidation or elimination of redundant or outdated applications to reduce costs and improve efficiency.

Traditional methods of application rationalization often involve manual audits and expert reviews. These approaches usually require extensive documentation and rely heavily on human judgment. While effective to some extent, they are time-consuming, costly, and subject to human error ([van Deursen et al., 2005](#)). Moreover, these methods often lack scalability and are not well-suited for large organizations with complex IT landscapes ([Ciborowska et al., 2021](#)).

With the advent of technology, automated approaches have started to gain traction. Tools and frameworks have been developed to assist in the rationalization process. One notable work in this area is by [Op 't Land et al. \(2023\)](#), who utilized Enterprise Ontology for rationalizing application portfolios at Rijkswaterstaat. The drawback, however, to this improvement over the traditional methods is its inapplicability in constantly changing environments which the subsurface application landscape is rife with.

Despite the availability of various tools and methodologies, many organizations still face challenges in effectively rationalizing their application landscapes. These challenges include data inconsistency, lack of interoperability, and the complexity of managing both legacy and modern applications ([Lee et al., 2022](#)). There is also a noticeable gap in the literature regarding the application of advanced computational methods, such as Natural Language Processing (NLP), in IT rationalization ([Pandita et al., 2021](#)).

Sentence similarity algorithms have become a cornerstone in the field of Natural Language Processing (NLP), offering a range of solutions for comparing textual data. These algorithms can broadly be categorized into basic and advanced types, each with its own set of applications and limitations.

Basic Algorithms

Cosine Similarity: This is a vector space model and is one of the earliest and most straightforward algorithms for measuring sentence similarity. It operates by calculating the cosine of the angle between two vectors, which are often term frequency-inverse document frequency (TF-IDF) representations of sentences. This algorithm has been widely applied in various domains such as document retrieval and text mining ([Wu and Palmer, 1994](#)). However, its simplicity comes at a cost. Cosine similarity is notably limited in its ability to capture semantic nuances between sentences, a limitation that has been highlighted in recent studies ([Zhang et al., 2018](#)).

Jaccard Index: This is another foundational algorithm, which measures the similarity between two sets by dividing the size of their intersection by the size of their union. It has found applications in clustering and information retrieval ([Hulth and Megyesi, 2006](#)). Despite its widespread use, the Jaccard index shares a common limitation with cosine similarity: it fails to capture the semantic meaning of sentences, a drawback that has been discussed in comparative studies ([Leskovec et al., 2007](#)).

Advanced Algorithms

Word2Vec maps words into a continuous vector space that captures syntax and semantics, enabling tasks like translation, sentiment analysis, and NER, but it typically needs large corpora to perform well ([Mikolov et al., 2013](#); [Ramesh et al., 2018](#)). **GloVe** blends global co-occurrence statistics with local context, often matching

Word2Vec’s effectiveness while being more computationally efficient in many settings (Pennington et al., 2014; Dey & Desai, 2019). BERT introduces bidirectional transformer context, setting strong benchmarks across diverse NLP tasks, yet its training and inference costs are substantial (Sanh et al., 2019; Tenney et al., 2019). Despite these advances, the literature offers little on applying sentence similarity methods to IT rationalization, leaving a clear opportunity to test these models for optimizing application landscapes.

Methodology

Data Collection

Our workflow for data preparation is shown above in Figure 1. We assembled a comprehensive dataset from an initial list of 149 subsurface applications by collecting descriptions from SMA-X pages, SharePoint, application databases, and validated vendor websites, with vendor and internal names obfuscated. Python pipelines automated extraction and normalization using BeautifulSoup for HTML, PyPDF2 for PDFs, and the SMA-X and SharePoint APIs, then consolidated all text into a CSV. Data quality was enforced through automated anomaly checks followed by expert manual review of a random sample. Each entry was annotated with metadata, including application name, discipline classification (for example Well Engineering, Exploration, Petroleum Engineering), and source of description, using automated tagging with expert validation. The resulting corpus varies in length and technical depth, yielding a diverse and representative foundation for subsequent analysis. An example table of the raw data is presented below in Table 1 and Table 2

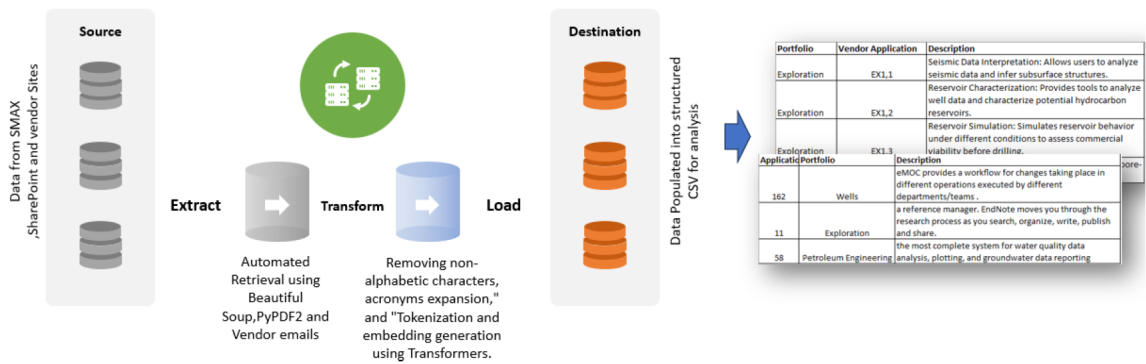


Figure 1—Data collection and preprocessing

Table 1—Raw Inhouse Data Format

Application Name	Discipline	Ontological Information	Source of Description	Application Description
Inhouse_1	Exploration	Client-server	Internal Repository	An integrated operational geology system, featuring Winlog, that manages geological workflows and integrates with drilling and other disciplines.
Inhouse_2	Well Engineering	Data Management	Service Management Automation X (SMA-X)	An integrated multi-user drilling and production engineering platform offering applications for well design, analysis, drilling, and completions.

Table 2—3rd Party Raw Data Sample

Application Name	Discipline	Ontological Information	Source of Description	Application Description
Application_1	Exploration	DM	Vendor website	A suite for seismic interpretation, reservoir characterization from well data, and reservoir simulation to delineate subsurface structures, assess hydrocarbon potential, and evaluate pre-drill commercial viability.

Preprocessing

A series of preprocessing steps were undertaken to standardize the text and improve the quality of the embeddings. The descriptions are concise narratives outlining the functionalities and characteristics of various applications used within our organization.

Data Cleaning: Application descriptions came from both manual and system sources, which introduced noise and inconsistency. We cleaned the text with regular expressions to remove special characters and numbers, then standardized terminology by expanding acronyms and abbreviations to their full forms using an internal glossary. The result is a cleaner, more uniform dataset that improves consistency across all application descriptions.

Tokenization and Stop Words Removal: Tokenization converts text into tokens for modeling. We use Sentence-Transformers all-MiniLM-L6-v2, whose BERT-based tokenizer performs subword tokenization so out-of-vocabulary terms are split into meaningful units, which helps with technical terms and acronyms; for company specific jargon, fine-tuning further improves coverage. Tokenization runs inside the model when text is provided, producing token and positional embeddings that preserve context for accurate sentence embeddings. We retain stop words because transformer models interpret full-sentence context without requiring their removal, which helps preserve the original meaning of the application descriptions.

Vectorization: Although an essential component in various NLP workflows, this step was not explicitly executed in our process, as the Sentence Transformers library seamlessly manages these intricacies internally.

Algorithm Selection

In resource-constrained, real-time settings such as on-prem data centers, **all-MiniLM-L6-v2** offers a strong balance of accuracy and efficiency. It has about **22.7M parameters** and encodes roughly **14,200 sentences per second**, enabling fast inference and modest memory use. This makes it well suited to computing semantic similarity between application descriptions, the core task for application landscape rationalization. In contrast, very large models like GPT-3.5 and GPT-4 have hundreds of billions of parameters, require specialized hardware, incur longer inference times, and are less practical to fine-tune on domain data. The smaller size of all-MiniLM-L6-v2 allows efficient domain fine-tuning, improving its understanding of organization-specific terminology while remaining deployable across diverse environments.

Experiment Design

Base Model Evaluation. In the preliminary phase, we encoded each application description with all-MiniLM-L6-v2 to produce high-dimensional embeddings that capture semantic nuance. We then computed pairwise cosine similarities to quantify relatedness across all descriptions. The resulting similarity matrix enabled systematic identification of semantically close applications, revealing functional overlap and potential redundancies. This analysis established a data-driven foundation for rationalizing the application landscape by highlighting candidates for consolidation and streamlining.

Fine-Tuning Process.

Data Generation Using GPT-4

The training dataset, pivotal to the fine-tuning of our language model, was uniquely generated using OpenAI's GPT-4, a language model known for its sophisticated text generation capabilities. The generation process was designed to create sentence pairs closely aligned with specific themes within the oil and gas sector, including exploration, well engineering, and petroleum engineering.

Prompt Design

To ensure technical accuracy and depth, we crafted a detailed prompt for GPT-4, requesting the creation of twenty pairs of intricate and specialized descriptions. Each pair was intended to reflect the complex nuances of the industry, mirroring real-world scenarios and terminologies. The prompt was strategically formulated to yield sentence pairs that were not only technically rich but also varied in terms of their semantic similarities.

Cosine Similarity Annotation

In addition to textual descriptions, each pair was accompanied by a cosine similarity score. These scores, represented as floating-point numbers, ranged from X to Y , where X is a float greater than or equal to 0 ($X \geq 0$), and Y is a float less than or equal to 1 ($Y \leq 1$). This range was chosen to quantify the semantic closeness of each sentence pair accurately, providing a robust basis for model training and evaluation. The scores were crucial for establishing ground truth labels, enabling the fine-tuning process to focus on learning the subtle degrees of similarity and dissimilarity within the context of the oil and gas sector.

The below was the prompt used:

"Craft twenty pairs of intricate and specialized descriptions pertinent to the oil and gas sector, encompassing areas such as exploration, well engineering, and petroleum engineering. Each pair should exhibit a nuanced technical depth. Accompany these pairs with a cosine similarity score, spanning a specified range from X to Y . Present these in a systematically organized table format for clarity and ease of analysis."

Fine-Tuning Methodology

We used 10-fold cross validation by splitting the dataset into ten equal parts, training on nine parts and testing on the remaining part in each iteration. This cycle ran ten times so every part served once as the test set. Averaging performance across folds provides a more comprehensive evaluation, reduces variance in estimates, and balances training and validation exposure to help mitigate overfitting.

Training Parameters

Model & Library: Sentence-Transformers with all-MiniLM-L6-v2, fine-tuned for semantic similarity in the oil & gas domain.

Loss Function: CosineSimilarityLoss to capture nuanced sentence-level similarity.

Batch Size: 50 — balances throughput and learning granularity.

Epochs: 10 per fold in cross-validation to learn sufficiently without overfitting.

Learning Rate: $2e-4$ for steady convergence.

Warmup: 10 steps to stabilize early training.

Evaluation Frequency: every 100 steps for ongoing performance monitoring and adjustments.

Dataset Structuring

The generated data was organized into a table format, enhancing clarity and facilitating ease of analysis. This structured approach was instrumental in streamlining the subsequent stages of data preprocessing and model training.

Selection for Training and Holdout

From the total of 525 generated sentence pairs, for selection of training and holdout sets, we employed a randomized approach to ensure unbiased distribution of data. Conceptually, this can be represented as:

$$S_i = \begin{cases} \text{Training Set if } Ri \leq 400/525 \\ \text{Holdout Set if } Ri > 400/525 \end{cases}$$

Where S_i represents the assignment of the i th sentence pair, and Ri is a random number between 0 and 1 generated for each pair. Sentence pairs were allocated to the training set if Ri was within the top 76% of the range (selecting 400 out of 525) and to the hold out set otherwise. This division was aimed at ensuring a comprehensive training process while reserving a significant portion of data for unbiased evaluation of the model's generalization capabilities.

Fine-Tuned Model Evaluation.

Process of Evaluating on the Holdout Set

We evaluated generalization on an unseen holdout set of 125 sentence pairs that were excluded from training to provide an unbiased assessment, detect overfitting, and gauge robustness. The fine-tuned model produced cosine similarity scores for each pair, which we compared with ground truth similarity labels. We then computed Mean Squared Error to quantify average prediction error and Spearman correlation to measure the strength of the monotonic relationship between predictions and labels, together indicating accuracy and semantic fidelity.

Comparison with Base Model

To quantify the effect of fine-tuning, we benchmarked the fine-tuned model against the base all-MiniLM-L6-v2 using Mean Squared Error (MSE) and Spearman correlation on a holdout set. MSE measures the average squared deviation between predicted and ground-truth similarity scores, so lower values indicate higher precision. Spearman correlation measures the strength and direction of the monotonic relationship between predictions and labels, which suits our non-parametric similarity data; higher values indicate better capture of semantic structure. Therefore, successful fine-tuning is evidenced by lower MSE and higher Spearman relative to the base model.

Model Averaging. Leveraging the hierarchical weight averaging approach validated by Gu et al. [24], we adopted ensemble modeling strategies to optimize predictive performance for energy industry challenges. Post k-fold cross-validation, systematic model averaging was implemented to eliminate fold-specific biases and enhance generalization capabilities, delivering the robust and reliable predictions required for critical petroleum engineering decision support systems.

$$\text{Average Weights } (W_{avg}) = \frac{1}{N} \sum_{i=1}^N W_i$$

Where:

W_{avg} is the average weights of the model after k-fold cross validation.

N is the number of folds in the cross-validation

W_i represents the weights of the model from the i^{th} fold.

Furthermore, by averaging, the final model encapsulates a more comprehensive learning from the entire dataset reducing variance in the model predictions ensuring $\text{Var}(W_{avg}) < \text{Var}(W_i)$

In addition, model averaging helps in stabilizing the predictions, making the final model more robust to variations in data, which is particularly crucial in real-world applications where data diversity is common. $\text{Std Dev}(\text{Predictions}|W_{avg}) < \text{Std Dev}(\text{Prediction}|W_i)$

Model Application for Rationalization. We extracted and preprocessed text descriptions from both Vendor and a range of Inhouse applications as shown in Table 1 and 2 to create out sentence pairs.

Descriptions=Extract (Vendor, Inhouse)

Embedding Generation: Using the fine-tuned model, these descriptions were converted into embeddings to capture their semantic content.

$$E_i = \text{Model}(\text{Description}_i)$$

For each Description_i , the model generates an embedding E_i . This embedding is a high-dimensional representation capturing the semantic content of the description.

Similarity Analysis: Cosine similarity was calculated between pairs of descriptions to quantify their semantic closeness.

$$\text{Sim}(E_i, E_j) = \frac{E_i \cdot E_j}{\|E_i\| \|E_j\|}$$

Visualization in Excel:

$$\text{Color}_{ij} = \text{MapColor}(E_i, E_j)$$

The similarity scores were translated into a color code, Color_{ij} , using a mapping function MapColor. This converts the similarity score into a corresponding color code, facilitating a visual representation of the data in an Excel sheet that highlights the relative similarity between vendor applications and in-house applications.

Overall Process:

$$\text{Rationalization} = \text{Visualize}(\text{Analyze}(\text{Embed}(\text{Preprocess}(\text{Descriptions}))))$$

This equation encapsulates the entire process, illustrating how each step build upon the previous one, starting from preprocessing and ending with a visual representation of the analysis.

Results and Discussion

Descriptive Statistics for Fine Tuning Data

This section presents a detailed statistical analysis of the dataset used for fine-tuning our sentence similarity model. We explore the length of application descriptions and their similarity scores to gain an in-depth understanding of the fine-tuning data characteristics. This analysis is crucial for assessing the data quality and its suitability for model optimization. Notably, a similar descriptive examination will be conducted with our case study data, allowing for a comprehensive comparison and deeper insights into model performance across different datasets.

Our analysis encompasses a dataset comprising 525 pairs of application descriptions, totaling 1,050 individual descriptions. This dataset is necessary for examining the similarity of applications using sentence similarity techniques. As discussed in the Prompt design section above, these descriptions were generated using GPT4, with generated sentence pairs with samples shown in [Table 3](#).

Table 3—Prompt output for similarity pairs

#	Description A	Description B	Similarity Score
1	Utilizing Fourier transform for seismic data interpretation in sub-salt regions.	Implementing sucker rod pumps for artificial lift in mature fields.	0.10
2	Conducting LWD for real-time reservoir navigation.	Designing gas lift systems for optimizing well performance.	0.15
3	Applying geostatistical models for stochastic reservoir simulation.	Using plunger lift mechanisms to manage liquid loading in gas wells.	0.12

Length Distribution of Descriptions

To understand the variability in description lengths, we examined the word count of each description. The lengths are distributed as shown in [Table 4](#).

Table 4—Description Statistics

Statistic	Description 1	Description 2
Mean Length	13.1 words	13.8 words
Standard Deviation	6.6 words	6.5 words
Range	5-37 words	6-34 words
Median	12 words	12 words

We further segmented these lengths into bins to provide a clearer picture of the distribution, as shown in Table 5. Examination of the results illustrates that most descriptions in both columns fall within the 5-10 and 11-15 word count ranges, indicating a tendency towards concise application descriptions.

Table 5.—Fine-Tune Data Word Count Statistics

Word Count Range	Description 1	Description 2
5-10	247	213
11-15	118	146
16-20	112	100
21-25	7	25
26-30	23	18
31-37	17	22

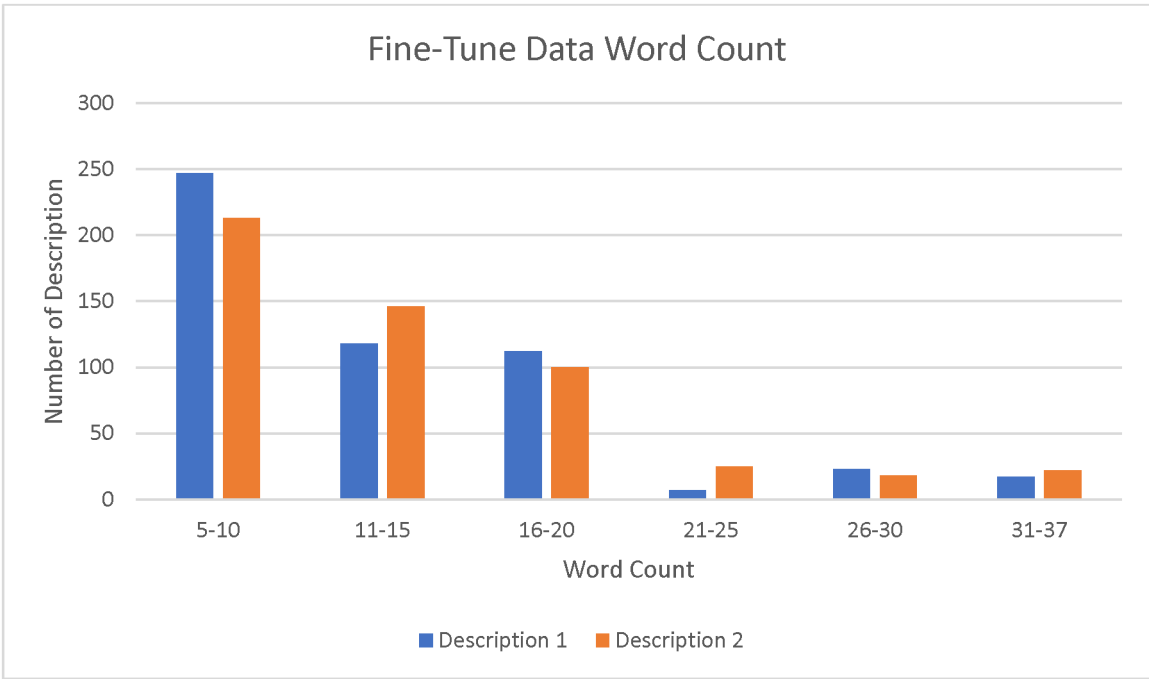


Figure 2—Distribution of Word Counts in Fine-Tune Descriptions

Similarity Score Distribution

The similarity between each pair of descriptions was quantified using a similarity score. The distribution of these scores is as follows:

Mean Similarity Score: Approximately 0.53

Standard Deviation: About 0.35
Range: 0.00 to 0.97
Median: 0.615

The mean similarity score of 0.53, coupled with a median of 0.615, indicates a moderate level of resemblance across the application descriptions in our dataset. This suggests a balanced mix of both closely related and distinct applications, underscoring the diversity of the dataset and highlighting its potential for a comprehensive evaluation of sentence similarity techniques.

Table 6 shows binned similarity into 5 bins, providing a more granular view of the distribution of similarities over the dataset.

Table 6—Similarity Ranges

Similarity Range	Number of Pairs
0 - 0.2	154
0.21 - 0.4	63
0.41 - 0.6	43
0.61 - 0.8	72
0.81 - 1	191

Evaluation Metrics

We employed two primary evaluation metrics to assess the performance of the fine-tuned ‘all-MiniLM-L6-v2’ algorithm in comparison to the base model: Mean Squared Error (MSE) and Spearman's Rank Correlation Coefficient. These metrics were chosen for their relevance and reliability in quantifying the accuracy and correlation effectiveness of the models.

Mean Squared Error (MSE)

Our analysis shows significant improvements in MSE across all folds after fine-tuning the model, refer to Figure 3. For instance, in Fold 1, the average MSE of the base model was 0.01804 compared to 0.01382 for the fine-tuned model(~25% reduction). Across all folds we notice a consistent decrease in cumulative MSE with Fold 10, achieving a Cumulative MSE of 0.00131 in the 10th epoch compared to a base model cumulative MSE of 0.02961(~95% reduction). Please see Appendix for complete table of results. The Holdout inference was done using the Averaged model discussed in the experimental section with a base model MSE of 0.0322 compared to 0.0160 for the finetuned model a 50% reduction in squared error for the model.

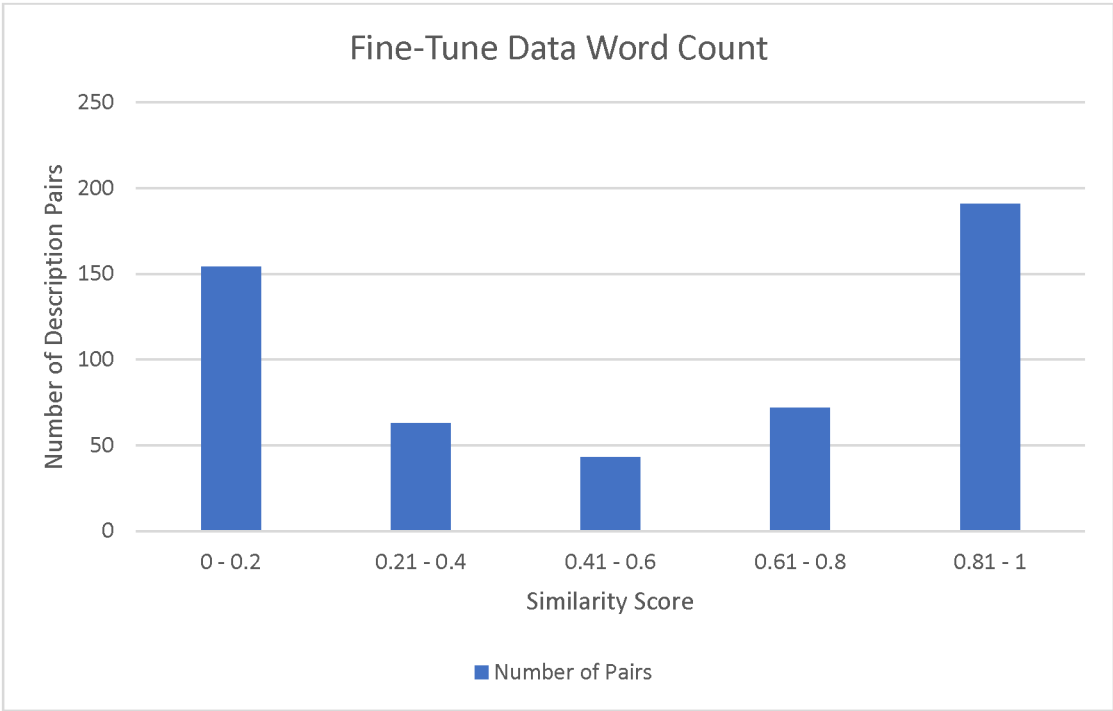


Figure 3—Histogram of Similarity Scores Between Sentence Pairs for Fine-Tuning.

Spearman's Rank Correlation Coefficient

The Spearman correlation analysis yielded substantial insights into the monotonic relationship strength between the predicted and actual values. As shown in Figure 4, the fine-tuned ‘all-MiniLM-L6-v2’ model demonstrated enhanced correlation coefficients across all folds except Fold 1, when compared to the base model.

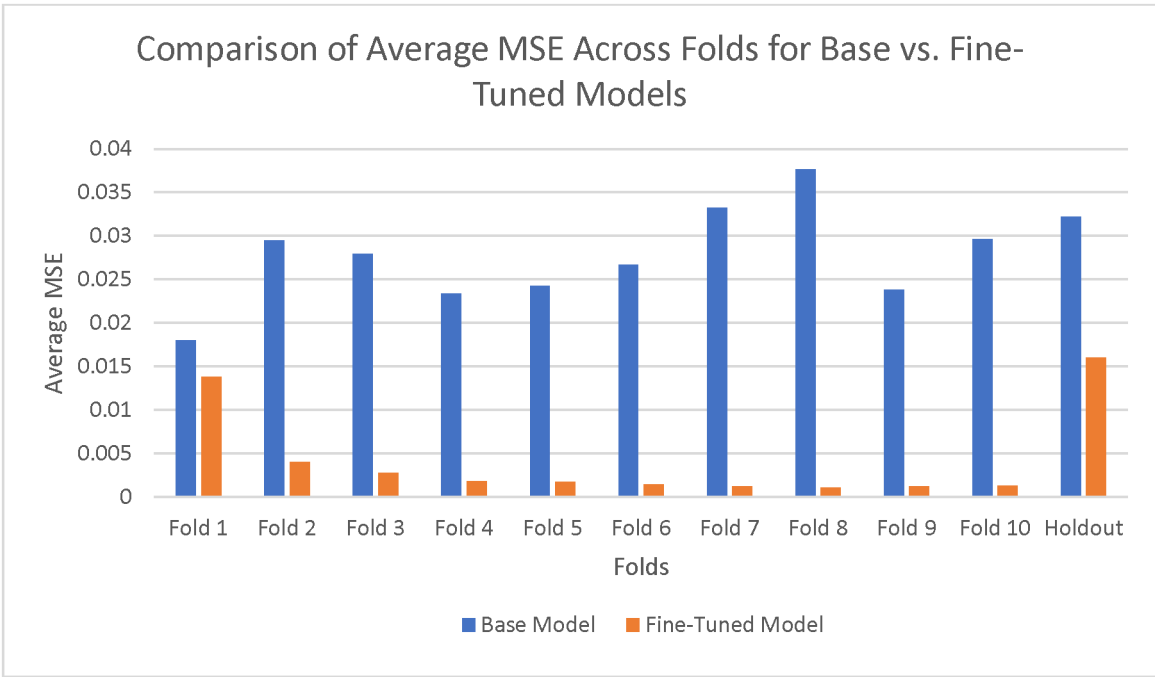


Figure 4—Comparison of Average MSE Across Folds for Base vs Fine-Tuned Models

Notably, Fold 2 exhibited a remarkable increase in Spearman correlation, increasing from 0.8504 in the base model to 0.9628 in the fine-tuned model, signifying a ~13% improvement in rank accuracy. This trend of enhanced monotonic relationships was consistent, with Fold 3 showing a Spearman correlation of 0.9693 vs 0.8140 for the base model, and Fold 10 achieving the highest Spearman correlation of 0.9911 vs 0.8724 for the base model, suggesting a strong and accurate ranking capability of the fine-tuned model.

The holdout set, used to validate the robustness of the model, which was an average of all weights from each fold, also reflected an improvement, with the fine-tuned model reaching a Spearman correlation of 0.8739, a notable increase from the base model's 0.8175. This denotes a significant enhancement in the model's ability to generalize beyond the training data, which is crucial for real-world application rationalization tasks.

Cumulatively, these results underscore the effectiveness of the fine-tuning process, not merely in reducing prediction error, as previously discussed in the context of MSE, but also in capturing the true ordering and ranking inherent in the data. It is evident from the analysis that fine-tuning has led to a model that is not only more precise but also more aligned with the actual data distribution, a critical factor for success in application rationalization efforts.

For a detailed account of the Spearman correlation coefficients across all folds, refer to [Appendix](#).

Case Study

In the dynamic and resource-intensive oil and gas industry, the strategic rationalization of applications is not merely a matter of IT efficiency but a cornerstone for competitive advantage. This case study presents a real-world application of our fine-tuned AI model, showcasing its potential to guide decision-makers in optimizing their application landscape.

Our analysis encompassed a comprehensive comparison of 149 in-house applications against 22 vendor applications on the latest cloud platform for subsurface data management. By quantitatively assessing the descriptions' similarity through our model, we established a correlation matrix that serves as a decision-support tool, revealing the potential for application consolidation, upgrade, or replacement.

The correlation matrix was segmented into bins reflecting similarity ranges, with each bin associated with strategic actions. Refer to [Figure 5](#) and [Figure 6](#):

- **High Compatibility (0.8 to 1.0):** Applications falling within this bin are prime candidates for replacement. The two applications identified here suggest a near-perfect match with vendor offerings, highlighting opportunities for IT modernization and cost rationalization.
- **Technical Evaluation Zone (0.6 to 0.79):** The 34 applications in this category warrant a technical evaluation. Discussions with subject matter experts (SMEs) are advised to delve into the nuances that the similarity scores may not fully capture, such as specific functional requirements or integration complexities.
- **Potential Overlook (0.4 to 0.59):** Dominating the analysis with 85 applications, this bin signifies moderate correlation. It suggests a need for a thorough review to ascertain if the in-house applications' descriptions are adequately detailed, as the AI model's performance is intrinsically linked to the quality of textual data.
- **Low Compatibility (0.39 and below):** Applications with a similarity score below 0.4, totaling 28, likely have unique features not mirrored by the vendor's platform. These applications might embody specialized processes crucial to the company's operations and hence are less likely to be replaced based on current market offerings.

The practical relevance of our research is underscored by its direct application in evaluating the application portfolio of a major player in the oil and gas industry. The insights derived from our AI model have illuminated pathways to not only enhance operational efficiency but also to harness the full potential of

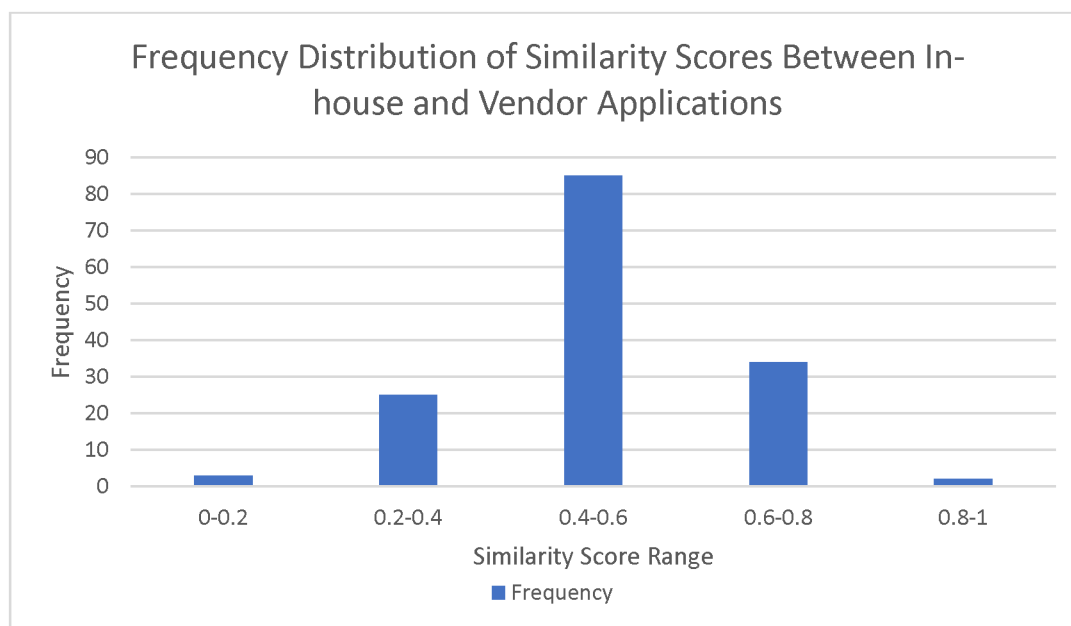


Figure 7—Frequency Distribution of Similarity Scores Between In-house and Vendor Applications

Discussion

The observed enhancements in model performance, as evidenced by the reduced MSE and increased Spearman correlation coefficients, suggest a substantial refinement in predictive accuracy and ranking consistency post fine-tuning. The fine-tuned ‘all-MiniLM-L6-v2’ algorithm demonstrates an ability to more closely mirror the actual patterns present within the data, an attribute that is important in the data-driven oil and gas sector where decision-making often relies on precise and robust predictive analytics.

These findings are in concordance with existing research (Kasai et al) that underscores the potential of fine-tuning transformer-based models for specialized tasks. Studies have consistently shown that fine-tuning can lead to significant gains in performance metrics, especially in domain-specific applications. Our research contributes to this body of knowledge by providing concrete evidence of such improvements within the context of application rationalization in the oil and gas industry, thereby bolstering the case for adopting advanced AI techniques in this field.

For the oil and gas industry, where operational efficiency, safety, and strategic planning are critical, the improved model's precision and reliability in predictive tasks could translate into more effective identification of rationalization opportunities. This could mean better capital allocation, more efficient maintenance scheduling, and optimized resource management, leading to cost savings and operational excellence while improving ROI on IT investments.

The slight decrease in the Spearman correlation coefficient for Fold 1 may indicate unique data characteristics better captured by the base model, or it could suggest that further model refinement is necessary to accommodate specific dataset nuances.

It is also worth noting that while the fine-tuned model excelled across most folds, the degree of improvement varied. This variability could be attributed to the inherent differences in data characteristics across folds or could suggest an area for further refinement of the fine-tuning process, ensuring even greater consistency in performance.

In conclusion, the empirical improvements observed through this study underscore the value of model fine-tuning and present a compelling case for the oil and gas industry to adopt such AI-driven methodologies. However, the noted anomalies also highlight the importance of a tailored approach, suggesting that fine-tuning should be considerate of the diverse data landscapes typical of this industry. Future research could build upon these findings by exploring the fine-tuning process in greater depth,

potentially leading to models that are even more attuned to the specific challenges and opportunities present in the oil and gas sector.

Conclusion

Our study has demonstrated the impactful application of fine-tuning a transformer-based model to enhance sentence similarity tasks within the oil and gas industry. By analyzing the model's performance across various datasets, we have identified significant improvements in predictive accuracy, as reflected by the reduced Mean Squared Error (MSE) and increased Spearman's Rank Correlation Coefficient. These advancements not only signify a technical achievement but also pave the way for practical, data-driven decision-making that can revolutionize application rationalization and operational efficiency.

The fine-tuned 'all-MiniLM-L6-v2' model's enhanced ability to discern nuanced textual similarities offers a robust tool for the industry, aiding in the identification of obsolete applications, the optimization of resource allocation, and the mitigation of operational risks. The case studies conducted across Exploration, Well Engineering, and Petroleum Engineering domains further solidify the model's utility and adaptability to the sector's diverse analytical needs.

Collaboration with domain experts will be essential to tailor the model to specific operational contexts, ensuring that the model's predictive power translates into tangible business outcomes. Future research should focus on refining the model's precision, expanding its scope to encompass broader datasets, and exploring its scalability across other sectors where data-driven insights can yield a competitive edge.

Limitations of the Study

While our study has yielded promising results, it is important to acknowledge its limitations. Firstly, the fine-tuning process and the subsequent evaluation were conducted on a dataset that, although diverse, may not capture the full spectrum of variability found in real-world applications. The linguistic nuances and technical jargon inherent to the oil and gas industry may require a larger and more varied dataset to fine-tune the model comprehensively.

Secondly, the model's reliance on textual descriptions means that the quality of input data is critical. In instances where descriptions are incomplete or lack specificity, the model's performance may be compromised. This highlights the need for consistently high-quality data curation practices.

Moreover, our study's scope was confined to the English language, which may not reflect the multilingual reality of global operations in the oil and gas sector. Future iterations of this work could extend to other languages and dialects to ensure broader applicability.

Finally, the rapidly evolving landscape of machine learning and AI technologies means that our model, while state-of-the-art at the time of the study, may need to be re-evaluated and updated as new advancements emerge. Keeping pace with technological progress will be vital to maintaining the model's relevance and accuracy.

In recognizing these limitations, we lay the groundwork for future research endeavors to build upon our findings, address these gaps, and further refine the model's capabilities.

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Appendices

1. Fine-Tuning Training results

Fold	Epoch	Base_Model MSE	Base_Model_Spearman Corr.	Fine_Tuned Model_MSE	Fine_Tuned Model_Spearman Corr.
1	1	0.0180	0.8683	0.0081	0.8685
1	2	0.0180	0.8683	0.0140	0.8377
1	3	0.0180	0.8683	0.0167	0.8427
1	4	0.0180	0.8683	0.0155	0.8395
1	5	0.0180	0.8683	0.0141	0.8392
1	6	0.0180	0.8683	0.0137	0.8408
1	7	0.0180	0.8683	0.0144	0.8492
1	8	0.0180	0.8683	0.0143	0.8501
1	9	0.0180	0.8683	0.0138	0.8501
1	10	0.0180	0.8683	0.0135	0.8519
2	1	0.0295	0.8504	0.0033	0.9750
2	2	0.0295	0.8504	0.0031	0.9765
2	3	0.0295	0.8504	0.0039	0.9577
2	4	0.0295	0.8504	0.0044	0.9530
2	5	0.0295	0.8504	0.0035	0.9566
2	6	0.0295	0.8504	0.0057	0.9552
2	7	0.0295	0.8504	0.0045	0.9668
2	8	0.0295	0.8504	0.0039	0.9668
2	9	0.0295	0.8504	0.0041	0.9603
2	10	0.0295	0.8504	0.0044	0.9606
3	1	0.0280	0.8140	0.0023	0.9669
3	2	0.0280	0.8140	0.0034	0.9642
3	3	0.0280	0.8140	0.0047	0.9688
3	4	0.0280	0.8140	0.0026	0.9673
3	5	0.0280	0.8140	0.0034	0.9572
3	6	0.0280	0.8140	0.0027	0.9655
3	7	0.0280	0.8140	0.0024	0.9741
3	8	0.0280	0.8140	0.0023	0.9736
3	9	0.0280	0.8140	0.0021	0.9776
3	10	0.0280	0.8140	0.0021	0.9779
4	1	0.0234	0.8566	0.0012	0.9603
4	2	0.0234	0.8566	0.0019	0.9470
4	3	0.0234	0.8566	0.0024	0.9482
4	4	0.0234	0.8566	0.0024	0.9507
4	5	0.0234	0.8566	0.0027	0.9506
4	6	0.0234	0.8566	0.0014	0.9532

Fold	Epoch	Base_Model MSE	Base_Model_Spearman Corr.	Fine_Tuned Model_MSE	Fine_Tuned Model_Spearman Corr.
4	7	0.0234	0.8566	0.0014	0.9557
4	8	0.0234	0.8566	0.0015	0.9521
4	9	0.0234	0.8566	0.0017	0.9498
4	10	0.0234	0.8566	0.0016	0.9484
5	1	0.0243	0.8697	0.0016	0.9869
5	2	0.0243	0.8697	0.0023	0.9793
5	3	0.0243	0.8697	0.0019	0.9753
5	4	0.0243	0.8697	0.0019	0.9818
5	5	0.0243	0.8697	0.0020	0.9793
5	6	0.0243	0.8697	0.0017	0.9821
5	7	0.0243	0.8697	0.0017	0.9852
5	8	0.0243	0.8697	0.0014	0.9867
5	9	0.0243	0.8697	0.0014	0.9897
5	10	0.0243	0.8697	0.0014	0.9881
6	1	0.0267	0.8761	0.0020	0.9873
6	2	0.0267	0.8761	0.0015	0.9834
6	3	0.0267	0.8761	0.0020	0.9783
6	4	0.0267	0.8761	0.0016	0.9833
6	5	0.0267	0.8761	0.0018	0.9780
6	6	0.0267	0.8761	0.0014	0.9802
6	7	0.0267	0.8761	0.0013	0.9772
6	8	0.0267	0.8761	0.0010	0.9833
6	9	0.0267	0.8761	0.0010	0.9862
6	10	0.0267	0.8761	0.0010	0.9855
7	1	0.0333	0.8224	0.0012	0.9920
7	2	0.0333	0.8224	0.0013	0.9864
7	3	0.0333	0.8224	0.0016	0.9892
7	4	0.0333	0.8224	0.0012	0.9838
7	5	0.0333	0.8224	0.0013	0.9764
7	6	0.0333	0.8224	0.0014	0.9807
7	7	0.0333	0.8224	0.0011	0.9791
7	8	0.0333	0.8224	0.0012	0.9828
7	9	0.0333	0.8224	0.0011	0.9818
7	10	0.0333	0.8224	0.0010	0.9834
8	1	0.0377	0.8312	0.0014	0.9894
8	2	0.0377	0.8312	0.0013	0.9837
8	3	0.0377	0.8312	0.0015	0.9708
8	4	0.0377	0.8312	0.0015	0.9758
8	5	0.0377	0.8312	0.0011	0.9853
8	6	0.0377	0.8312	0.0008	0.9870

Fold	Epoch	Base_Model MSE	Base_Model_Spearman Corr.	Fine_Tuned Model_MSE	Fine_Tuned Model_Spearman Corr.
8	7	0.0377	0.8312	0.0008	0.9915
8	8	0.0377	0.8312	0.0008	0.9886
8	9	0.0377	0.8312	0.0008	0.9854
8	10	0.0377	0.8312	0.0008	0.9858
9	1	0.0238	0.8810	0.0011	0.9822
9	2	0.0238	0.8810	0.0018	0.9822
9	3	0.0238	0.8810	0.0024	0.9744
9	4	0.0238	0.8810	0.0015	0.9806
9	5	0.0238	0.8810	0.0011	0.9806
9	6	0.0238	0.8810	0.0009	0.9777
9	7	0.0238	0.8810	0.0010	0.9809
9	8	0.0238	0.8810	0.0009	0.9841
9	9	0.0238	0.8810	0.0009	0.9838
9	10	0.0238	0.8810	0.0009	0.9837
10	1	0.0296	0.8724	0.0007	0.9945
10	2	0.0296	0.8724	0.0020	0.9904
10	3	0.0296	0.8724	0.0017	0.9928
10	4	0.0296	0.8724	0.0014	0.9898
10	5	0.0296	0.8724	0.0014	0.9919
10	6	0.0296	0.8724	0.0015	0.9905
10	7	0.0296	0.8724	0.0011	0.9905
10	8	0.0296	0.8724	0.0012	0.9895
10	9	0.0296	0.8724	0.0011	0.9903
10	10	0.0296	0.8724	0.0010	0.9918
Holdout	NA	0.0322	0.0818	0.0160	0.8739